THE UNIVERSITY OF ARIZONA
COLLEGE OF MEDICINE PHOENIX
Anesthesiology

CEREBROSPINAL FLUID ANALYSIS ALGORITHM

Interpretation and Diagnostic Approach for the Neurocritical Care Patient

Purpose

To provide a structured, stepwise approach to ordering and interpreting cerebrospinal fluid (CSF) studies in the neurocritical care patient, distinguishing bacterial, viral, fungal/tuberculous, and hemorrhagic etiologies, and guiding reflex/confirmatory testing and empiric management decisions.

Normal CSF Reference Values

Parameter	Normal Adult Value
Opening pressure	10–20 cmH2O (lateral decubitus, legs extended)
Appearance	Clear, colorless
WBC count	≤5 cells/mm³ (mononuclear predominant)
RBC count	0 (any RBCs raise question of traumatic tap vs hemorrhage)
Protein	15–45 mg/dL (higher in elderly, lumbar > cisternal > ventricular)
Glucose	45–80 mg/dL; CSF:serum glucose ratio ≥0.6

Step 1 — Before the Tap

- Assess for contraindications: signs of elevated ICP with risk of herniation, coagulopathy/thrombocytopenia, overlying skin infection, spinal epidural abscess concern.
- Obtain CT head first if: age >60, immunocompromised, history of CNS disease, new seizure within 1 week, altered consciousness, or focal neurologic exam — per standard practice, to assess for mass effect before lumbar puncture.
- Do not delay empiric antibiotics/antivirals for suspected bacterial or HSV meningoencephalitis while awaiting imaging or LP — obtain blood cultures first and treat empirically.

Step 2 — Initial CSF Studies (Send on Every Tap)

- Opening pressure (lateral decubitus, legs extended, patient relaxed)
- Tube 1: Cell count and differential
- Tube 2: Glucose and protein (paired serum glucose drawn at the same time)
- Tube 3: Gram stain and bacterial culture
- Tube 4: Cell count and differential (repeat — to compare with tube 1 if traumatic tap or SAH is a concern), hold for additional studies (HSV/VZV/enterovirus PCR, cryptococcal antigen, cytology, AFB, VDRL, oligoclonal bands) as clinically indicated

Step 3 — Interpret the Pattern

Compare the CSF profile against the etiology-specific patterns below to guide the differential and next steps. No single parameter is fully sensitive or specific — interpret the full picture together with the clinical context.

Parameter	Bacterial	Viral / Aseptic	Fungal / TB	SAH
Opening pressure	Often elevated, 200–500 mmH2O	Normal to mildly elevated	Often elevated	Elevated
WBC count / differential	1,000–5,000/mm³ (range <100 to >10,000); neutrophil predominant (80–95%)	10–500/mm³; lymphocyte predominant (may be neutrophil-predominant in first 24 h)	10–500/mm³; lymphocyte predominant, subacute/chronic course	RBCs markedly elevated; WBC:RBC ratio approximates peripheral blood ratio (~1:500–1,000)
Protein	Elevated, often >250 mg/dL	Mildly elevated, 50–100 mg/dL	Elevated, often markedly high in TB	Elevated (blood breakdown products)
Glucose	Low, <40 mg/dL; CSF:serum ratio ≤0.4 (80% sens / 98% spec)	Normal	Low-normal; often low in TB	Normal
Confirmatory studies	Gram stain (60–90% sensitive untreated); culture; consider CSF lactate if post-neurosurgical	HSV/VZV/enterovirus PCR; multiplex meningitis/encephalitis PCR panel	Cryptococcal antigen, India ink, fungal culture; AFB smear/culture and/or TB PCR (low sensitivity — send large volume, repeat if suspicion high)	Xanthochromia (develops ~12 h post-bleed); RBC count does not clear between tube 1 and tube 4

Step 4 — Reflex and Confirmatory Testing by Suspected Category

Suspected Bacterial Meningitis

- Gram stain and culture on all patients with suspected meningitis; start empiric antibiotics ± dexamethasone without delay.
- If CSF to serum glucose ratio and Gram stain are equivocal, consider CSF lactate (particularly useful in post-neurosurgical/EVD patients, where standard indices are less reliable).
- PCR/multiplex panels can identify a pathogen when cultures are negative due to prior antibiotic exposure.

Suspected Viral Meningitis/Encephalitis

- Send HSV-1/2 PCR on all patients with encephalitis or unexplained aseptic meningitis — start empiric IV acyclovir while results are pending given the consequences of untreated HSV encephalitis.
- Enterovirus PCR, VZV PCR, and a multiplex meningitis/encephalitis panel as available and clinically indicated.
- A negative HSV PCR very early in illness (<72 h of symptom onset) does not fully exclude HSV encephalitis — repeat if clinical suspicion remains high.

Suspected Fungal or Tuberculous Meningitis

- Cryptococcal antigen (CrAg) and India ink in immunocompromised patients (very high sensitivity for CrAg).
- Send a large CSF volume (≥10 mL if possible) for AFB smear/culture and/or TB PCR — AFB smear sensitivity is low; repeat sampling improves yield.
- Consider fungal culture and beta-D-glucan in the appropriate host.

Suspected Subarachnoid Hemorrhage / Traumatic Tap

- Xanthochromia (visual or spectrophotometric) develops over several hours and is most reliable ≥12 hours after symptom onset — a negative CT plus negative xanthochromia effectively excludes SAH in this window.

- RBC counts that fall substantially between tube 1 and tube 4 favor a traumatic tap over true SAH; RBCs that remain similar across tubes favor SAH.
- Corrected WBC for a traumatic tap: subtract roughly 1 WBC for every 500–1,000 RBCs present, adjusted for the patient's peripheral WBC:RBC ratio when known, rather than applying a fixed rule to every patient.

Malignancy / Carcinomatous Meningitis

- Send cytology on at least 10–15 mL of CSF; sensitivity of a single sample is only ~50–60% — repeat sampling (up to 3 taps) substantially increases yield.
- Flow cytometry if a hematologic malignancy is suspected.

Special Populations — Interpret With Caution

• Post-neurosurgical / EVD / CSF shunt patients: standard cell count, glucose, and protein thresholds are unreliable for distinguishing infection from sterile post-operative inflammation. CSF lactate and trends over time, combined with clinical status and repeat cultures, are more useful. Diagnosis often requires a combination of pleocytosis, positive culture (ideally on ≥2 samples), and clinical findings rather than any single CSF value.
• Immunocompromised patients: classic CSF abnormalities may be blunted or absent despite significant CNS infection — maintain a low threshold for expanded testing (CrAg, AFB/TB PCR, viral PCR panel, cytology) even with a relatively unremarkable initial profile.
• Partially treated bacterial meningitis (antibiotics prior to LP): Gram stain and culture yield drop substantially; rely more heavily on PCR-based testing and the overall clinical/CSF chemistry picture.

When the CSF picture and clinical presentation are discordant, treat empirically for the most dangerous plausible diagnosis while confirmatory testing is pending.

References

1. Tunkel AR, Hartman BJ, Kaplan SL, et al. Practice Guidelines for the Management of Bacterial Meningitis. Clin Infect Dis. 2004;39(9):1267–1284. doi:10.1086/425368. PMID: 15494903.

2. Tunkel AR, Hasbun R, Bhimraj A, et al. 2017 Infectious Diseases Society of America's Clinical Practice Guidelines for Healthcare-Associated Ventriculitis and Meningitis. Clin Infect Dis. 2017;64(6):e34–e65. doi:10.1093/cid/ciw861. PMID: 28203777.